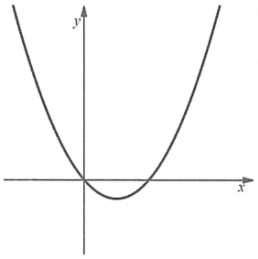
Differentiation: Graphs of Gradient Functions

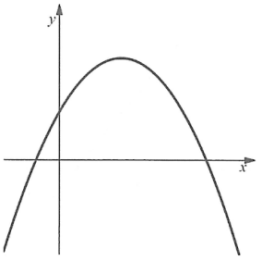
Questions 1–4

Question 1. Draw the gradient function on copies of each of the following graphs:

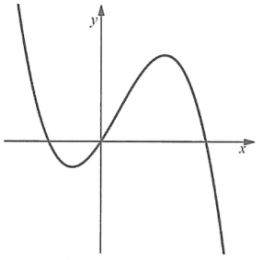
a)



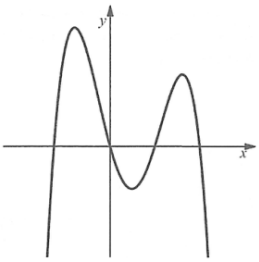
b)



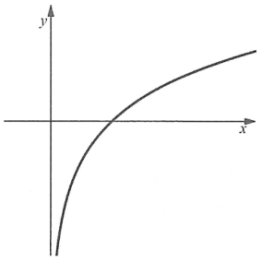
c)



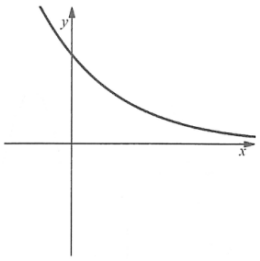
d)



e)

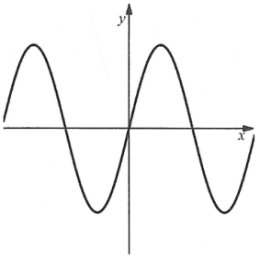


f)

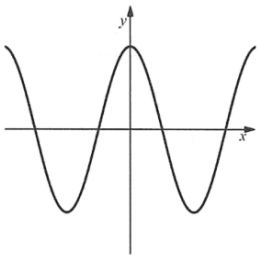


Question 2. Given that in each of the following functions, their maximum and minimum gradient is the same magnitude as the maximum and minimum values respectively, draw the gradient function on copies of each of the following graphs:

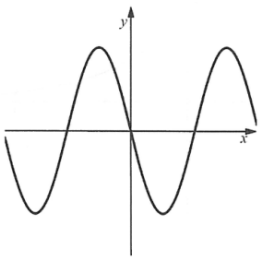
a)



b)



c)

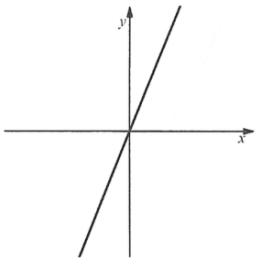


Question 3. The gradient function of the curve

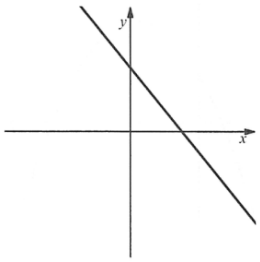
$$f(x) = ax^3 + bx^2 + cx$$

is sketched.
In each case state whether each of a, b, and c are greater than, equal to or less than 0

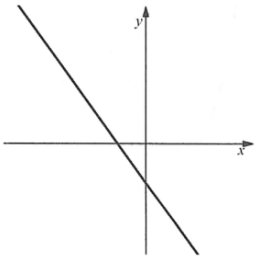
a)



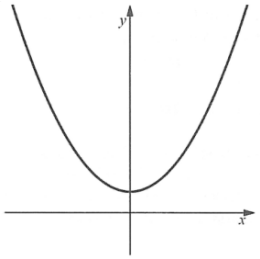
b)



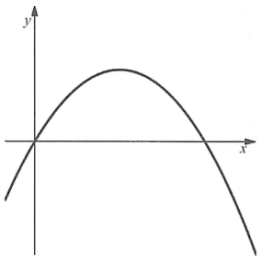
c)



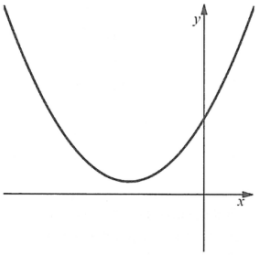
d)



e)



f)

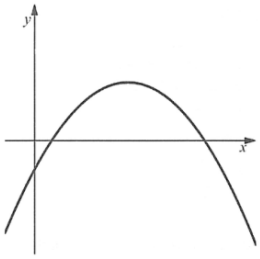


Question 3 (continued). The gradient function of the curve

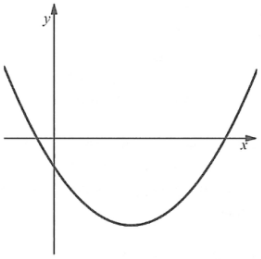
$$f(x) = ax^3 + bx^2 + cx$$

is sketched.
In each case state whether each of a, b, and c are greater than, equal to or less than 0

g)



h)



Question 4. For the curve shown on the graph below, the two points of inflection are labelled A and B. Sketch the gradient function on a copy of the graph.

